



Natural Language Processing

Exercise Sheet 1: NLP Problems

Exercise 1

Think of an application of natural language processing that did **not** appear in the lecture slides and briefly describe it. Ideally, choose an application that your fellow students are unlikely to have thought of.

Exercise 2

For the paragraph below, referring to the examples shown in the lecture slides, do the following **by hand**:

- a) Tokenisation
- b) Lemmatisation
- c) POS tagging
- d) Named Entity Recognition
- e) Identify all coreferences
- f) Entity relation extraction
- g) Count unigrams, bigrams, and trigrams and create a frequency list of each

What were challenging cases where you had to make a decision, and how did you make it?

Paragraph:

The wise Sir Bedemir was the first to join King Arthur's knights, but other illustrious names were soon to follow: Sir Lancelot the Brave; Sir Galahad the Pure; and Sir Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of Angnor, who had nearly stood up to the vicious Chicken of Bristol and who had personally wet himself at the Battle of Badon Hill; and the aptly named Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries, the Knights of the Round Table.